From Text to Signals in Finance

Lecture 1 — Introduction

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Outline

The Founding Moment: Tetlock (2007)

The Setup

Autumn 2006. Paul Tetlock assembles a dataset:

- Daily fraction of negative words in the *WSJ* "Abreast of the Market" column (1984–1999)
- Next-day Dow Jones Industrial Average return

Results:

- High negativity ⇒ lower next-day returns [puzzling under EMH]
- Effect *reverses* over the subsequent week ⇒ temporary mispricing
- High/low negativity ⇒ high next-day trading volume

THE BIGGER PICTURE

Human language can be turned into a quantitative return-predicting signal. The reversal tells us it reflects sentiment, not fundamental news.

Dictionary Methods: Harvard IV-4 and LM

General Inquirer (Stone et al., 1966)

- 11,000 word stems classified into Positive, Negative, Strong, Weak, . . .
- Score a document: count words per category, normalise by length
- Transparent and reproducible

Loughran–McDonald (LM, 2011)

- *tax*, *liability*, *depreciation* flagged “negative” by Harvard IV-4 but are merely financial jargon
- LM dictionary hand-coded from 10-K filings: Negative, Positive, Uncertainty, Litigious, Strong Modal, Weak Modal
- Substantially outperforms Harvard IV-4 in predicting filing-date abnormal returns

Fundamental limitation of all dictionary methods

Static, order-invariant. “Earnings did *not* decline” = “Earnings *did* decline” unless negation is hard-coded.

Sources of Financial Text

Earnings calls Scripted presentation $+$ unscripted Q&A; Q&A is spontaneous and more information-rich

10-K / 10-Q filings MD&A, risk factors; up to 200 pages; primary source for Loughran–McDonald (2011)

Newswire / news High frequency; narrow exploitation window; value in *speed* and *breadth*

Analyst reports Informationally dense; linguistic complexity $\propto$ information asymmetry

Central bank comms FOMC minutes and statements move rates and equity prices on release

Social media High-frequency, noisy; predictive power documented but robustness debated

Why Text Signals Survive Market Efficiency

Semi-strong EMH (Fama, 1970): prices reflect all publicly available information. Yet large empirical literature finds text predicts returns. Three reconciling mechanisms:

1. **Processing costs & limited attention**
   A 200-page 10-K is costly to read. Not all investors process it fully. NLP at scale makes it feasible $\Rightarrow$ anomaly should attenuate with technology.
2. **Soft information & interpretive disagreement**
   Cautious tone = genuine uncertainty *or* expectation management? Disagreement $\Rightarrow$ prices reflect average of beliefs, not the most informative one.
3. **Latent information**
   Regulation FD limits explicit selective disclosure. Information leaks through word choice, omissions, and rhetorical structure.

THE BIGGER PICTURE

Text analysis adds value not because markets are irrational, but because

Four Empirical Regularities

The theory must explain these stylised facts:

1. **Negative tone $\Rightarrow$ negative short-run returns, with reversal.** Tetlock (2007): pessimistic WSJ prose $\rightarrow$ lower next-day DJIA, then mean-reversion.
2. **Financial-domain dictionaries outperform general-domain ones.** Loughran–McDonald (2011): LM Negative $\gg$ Harvard IV-4 for predicting filing-date abnormal returns.
3. **Earnings call tone predicts future earnings surprises.** Larcker–Zakolyukina (2012): deceptive CEO/CFO speech has distinct linguistic markers.
4. **LLM signals show incremental predictive power.** Lopez-Lira & Tang (2023): ChatGPT sentiment for headlines predicts next-day returns; absent for GPT-1 and BERT $\Rightarrow$ emergent property of scale.

The Textual Signal: Formal Definition

Definition

A **textual signal** is a measurable function

$$f : \mathcal{D} \longrightarrow \mathbb{R}^k$$

mapping document d_t (at time t) to signal value $s_t = f(d_t)$.

Examples of f:

- Fraction of LM Negative words ($k = 1$)
- Vector of LDA topic proportions ($k =$ number of topics)
- Word2Vec sentence embedding ($k = 300$)
- LLM classifier output ($k =$ number of classes)

This course = investigating the best choice of f: from simple word counts → topic models → transformer embeddings → LLMs as reasoning agents.

The Linear Factor Model and Economic Significance

The central empirical question:

$$\mathrm{Cov}(s_t,\ r_{t+1}) \neq 0\ ?$$

Under a linear factor model:

$$r_{t+1} = \alpha + \beta^\top s_t + \varepsilon_{t+1}, \qquad \mathbb{E}[\varepsilon_{t+1} \mid s_t] = 0$$

β is **economically significant** if and only if:

1. $|\beta| > 0$ after controlling for size, value, momentum, ...
2. The implied Sharpe ratio exceeds plausible transaction costs

Both conditions are necessary

Statistical significance without economic significance does not justify a trading strategy.

Identification Challenges

Showing $\mathrm{Cov}(s_t, r_{t+1}) \neq 0$ is necessary but *not sufficient*:

Reverse causality Document d_t may be written *in response to* lagged returns, not in anticipation of future ones.

Confounding Macro conditions drive both tone and returns. Use high-frequency event studies with known release times.

Multiple testing NLP produces thousands of candidate signals. Use strict temporal out-of-sample split (train pre-2010, test post-2010).

Model misspecification The choice of f is itself an assumption. Dictionary f assumes only word counts matter; LLM f relaxes this but introduces training-data biases.

Tokenisation and the Preprocessing Pipeline

Before computing any representation, raw text is standardised:

1. **Case folding** — "Revenue" $\equiv$ "revenue"
2. **Stop-word removal** — remove "the", "and", "of", ... (100–300 words)
3. **Stemming / lemmatisation**
   - Stemming: *earnings, earned, earns* $\to$ *earn*
   - Lemmatisation: *better* $\to$ *good* (more accurate, slower)
4. **Vocabulary truncation** — retain top V' types; discard very rare words

Result: finite vocabulary $\mathcal{V}$ of size V (typically 10 000–100 000)

Note: LLMs use *sub-word tokenisation* — "earnings" $\to$ ["earn", "ings"] — allowing rare words to be represented as combinations of frequent sub-units.

Bag-of-Words Model

Definition

$$\mathbf{x}(d) = \big(\mathrm{count}(v_1, d), \ldots, \mathrm{count}(v_V, d)\big)^\top \in \mathbb{Z}_{\geq 0}^{V}$$

Word order is discarded. The document is treated as an unordered multiset.

Virtues

- Simple and interpretable
- Sparse — efficient storage (CSR/CSC)
- Competitive baseline for many tasks

The fundamental deficiency

- "The fund outperformed the market" $=$ "The market outperformed the fund" (identical BoW vectors, opposite meanings)

The term-document matrix $\mathbf{X} \in \mathbb{Z}_{\geq 0}^{N \times V}$ stacks BoW vectors for all N documents.

One-Hot Encoding: The Discrete Metric Problem

Each token v_k maps to the standard basis vector:

$$\mathbf{e}_k = (\underbrace{0, \ldots, 0}_{k-1}, 1, \underbrace{0, \ldots, 0}_{V-k})^\top \in \{0, 1\}^V$$

BoW = sum of one-hot vectors of constituent tokens.

UNDER THE HOOD

The problem: orthogonality.

$\mathbf{e}_j^\top \mathbf{e}_k = \mathbb{1}[j = k] \quad \Rightarrow \quad \mathrm{sim}(\text{revenue, income}) = 0 = \mathrm{sim}(\text{revenue, pengui}$

Every pair of distinct words is equally dissimilar — the *discrete metric*. This is semantically incorrect: financial analysts know that "revenue" and "income" are nearly synonymous in most contexts.

Word embeddings replace the discrete metric with a continuous geometry that reflects distributional similarity.

TF-IDF: Motivation and Definitions

Raw BoW counts dominated by stop words. TF-IDF weights by document-level rarity:

$$\underbrace{\mathrm{tf}(v,d)}_{\text{term freq.}} = \frac{\mathrm{count}(v,d)}{|d|} \qquad \underbrace{\mathrm{idf}(v,\mathcal{C})}_{\text{inv. doc. freq.}} = \log\left(\frac{|\mathcal{C}|}{\mathrm{df}(v,\mathcal{C})}\right)$$

$$\mathrm{tfidf}(v,d,\mathcal{C}) = \mathrm{tf}(v,d) \; \times \; \mathrm{idf}(v,\mathcal{C})$$

Intuition of IDF:

- Word in *every* document $\Rightarrow \mathrm{df} = |\mathcal{C}| \Rightarrow \mathrm{idf} = \log(1) = 0$ (suppressed)
- Word in *one* document $\Rightarrow \mathrm{idf} = \log|\mathcal{C}|$ (maximally discriminative)

Document similarity: cosine on ℓ_2-normalised TF-IDF vectors

$$\mathrm{sim}(d_i, d_j) = \frac{\tilde{\mathbf{x}}(d_i)^\top \tilde{\mathbf{x}}(d_j)}{\|\tilde{\mathbf{x}}(d_i)\|_2 \, \|\tilde{\mathbf{x}}(d_j)\|_2} \; \in \; [0,1]$$

Limitations of Classical Methods

Despite TF-IDF's improvements over raw counts, all classical methods share:

1. **Order invariance**
   "Earnings beat estimates" = "Estimates beat earnings" (identical BoW/TF-IDF vectors)
2. **No semantic similarity**
   Synonyms that never co-occur (*revenue*, *sales*) have zero cosine similarity
3. **Fixed vocabulary**
   New words at test time (acronyms, company names) are out-of-vocabulary and discarded
4. **High-dimensional sparsity**
   $V \sim 10^4$–10^5; even with IDF weighting, unsuitable as direct inputs to dense neural layers without dimensionality reduction

Solution: replace the discrete one-hot basis with a continuous embedding space.

The Distributional Hypothesis

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"You shall know a word by the company it keeps." — J.R. Firth (1957)

Words with similar meanings appear in similar contexts. Capture contextual co-occurrence $\Rightarrow$ capture semantic similarity.

Definition: Word Embedding

An injective map $\varphi : \mathcal{V} \to \mathbb{R}^d$ with $d \ll V$.
Embedding matrix $\mathbf{W} \in \mathbb{R}^{V \times d}$; row $i = \varphi(w_i)$.
Typical: $d \in \{50, 100, 300\}$; compression ratio $V/d \approx 150$–1,000.

Geometry: under a well-trained embedding, cosine similarity $\approx$ semantic similarity.
Linear analogies: $\varphi(\text{king}) - \varphi(\text{man}) + \varphi(\text{woman}) \approx \varphi(\text{queen})$
In finance: $\varphi(\text{call option}) - \varphi(\text{upside}) + \varphi(\text{downside}) \approx \varphi(\text{put option})$

Word2Vec: Skip-Gram Objective

Each centre word w_t predicts context words in window $\pm c$:

$$J_{\mathrm{SG}} = \frac{1}{T}\sum_{t=1}^{T}\sum_{\substack{k=-c\\k\neq 0}}^{c}\log P(w_{t+k} \mid w_t)$$

where the softmax probability is:

$$P(w_{t+k} \mid w_t) = \frac{\exp\left(\mathbf{v}_{w_{t+k}}^{\top}\hat{\mathbf{v}}_{w_t}\right)}{\sum_{w=1}^{V}\exp\left(\mathbf{v}_{w}^{\top}\hat{\mathbf{v}}_{w_t}\right)}$$

UNDER THE HOOD

The computational problem. Normalising constant sums over all $V \approx 10^4$–10^5 words *at every gradient step*. For 1 billion tokens: intractable.

Two embedding matrices: *centre-word* vectors $\hat{\mathbf{v}}_w$ and *context-word* vectors $\mathbf{v}_w$.

Negative Sampling: Efficient Training

Replace the full softmax with a binary classification task:

- 1 *positive* sample: the observed context word w_{t+k}
- K *negative* samples: words drawn from noise distribution P_n

$$\mathcal{L}_{\mathrm{NEG}}(w_t, w_{t+k}) = \log \sigma\left(\mathbf{v}_{w_{t+k}}^\top \hat{\mathbf{v}}_{w_t}\right) + \sum_{j=1}^{K} \mathbb{E}_{w_j \sim P_n}\left[\log \sigma\left(-\mathbf{v}_{w_j}^\top \hat{\mathbf{v}}_{w_t}\right)\right]$$

Gradient interpretation:

$$\frac{\partial \mathcal{L}_{\mathrm{NEG}}}{\partial \mathbf{v}_{w_{t+k}}} = \left[1 - \sigma\left(\mathbf{v}_{w_{t+k}}^\top \hat{\mathbf{v}}_{w_t}\right)\right] \hat{\mathbf{v}}_{w_t}$$

Positive context vector is pushed *toward* centre vector (large coefficient when dissimilar); negative sample vectors are pushed *away*.

Efficiency: only $K+1$ rows of $\mathbf{W}$ touched per step, not all V.
Recommended: $K = 5$–20; $P_n(w) \propto \mathrm{freq}(w)^{3/4}$.

Finance-Specific Domain Adaptation

General embeddings (Wikipedia/Common Crawl) encode general-English meaning:

Word	General nearest neighbours	Finance nearest neighbours
risk	danger, threat, hazard	volatility, exposure, VaR
short	brief, concise	short-sell, bear position
margin	edge, border	profit margin, maintenance margin

Domain adaptation strategies:

1. **Pre-train from scratch** on financial corpus (Bloomberg Financial Word2Vec; FinancialPhraseBank embeddings)
2. **Fine-tune transformer embeddings** (FinBERT: BERT re-trained on Reuters news, 10-K filings, earnings calls)
3. **Retrofitting**: post-hoc adjustment using a financial lexicon or ontology to pull synonyms together and push antonyms apart

What We Have Covered and What Comes Next

This lecture:

- Why text predicts returns: three EMH-consistent mechanisms
- Formal textual signal framework; linear factor model; identification challenges
- Classical representations: BoW, TF-IDF, one-hot — order-invariant, discrete metric
- Word embeddings: distributional hypothesis, Word2Vec (NEG), GloVe, domain adaptation

Practical Session 1: TF-IDF by hand · cosine similarity · Python: TF-IDF on earnings call snippets **Lecture 2 (LLM Foundations):**

- RNNs: sequential order, vanishing gradients
- LSTM: cell state, forget/input/output gates
- Transformer: scaled dot-product attention, multi-head, positional encoding
- Pre-training objectives: masked LM (BERT) vs. autoregressive (GPT)

Key refs: Tetlock (2007) · LM (2011) · Mikolov (2013) · Pennington (2014) ·

Appendix

Extended Examples, Timeline, and GloVe

Extra / optional material — beyond the core lecture.

From Dictionaries to Neural Methods: A Timeline

Year	Method	Key advance
1966	General Inquirer	Systematic word counting
2003	LDA (topic models)	Data-driven topics, no labels needed
2011	LM Dictionary	Financial-domain sentiment
2013	Word2Vec / GloVe	Dense embeddings, semantic geometry
1997/2015	LSTM / Attention	Sequential context, negation
2017	Transformer	Parallel self-attention, massive scale
2018–19	BERT / FinBERT	Pre-train then fine-tune
2020+	GPT-4, Claude, Llama	Instruction-following, reasoning

Each generation addresses the core limitation of the previous one.
Understanding this history explains *why* LLMs behave the way they do.

TF-IDF: Worked Example

Three pre-processed financial sentences ($N = 3$, $V = 8$):

d	Tokens
d_1	`revenue grew strong earnings`
d_2	`earnings declined weak guidance`
d_3	`revenue guidance raised strong`

$\mathrm{idf}(\texttt{declined}) = \mathrm{idf}(\texttt{grew}) = \mathrm{idf}(\texttt{raised}) = \mathrm{idf}(\texttt{weak}) = \ln(3) \approx 1.099$
$\mathrm{idf}(\texttt{earnings}) = \mathrm{idf}(\texttt{guidance}) = \mathrm{idf}(\texttt{revenue}) = \mathrm{idf}(\texttt{strong}) = \ln(3/2) \approx 0.405$

Cosine similarities

$\mathrm{sim}(d_1, d_3) \approx 0.213$ (shared: `revenue, strong`)
$\mathrm{sim}(d_1, d_2) \approx 0.106$ (shared: `earnings` only)

Full computation in Practical Session 1.

GloVe: Global Co-occurrence Factorisation

Co-occurrence matrix X_{ij} = times word j appears in context of word i across the full corpus. **Key insight:** the *ratio* P_{ik}/P_{jk} encodes relational meaning more faithfully than raw probabilities. GloVe minimises:

$$J_{\mathrm{GloVe}} = \sum_{i,j=1}^{V} f(X_{ij}) \left(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

with weighting function $f(x) = (x/x_{\mathsf{max}})^\alpha$ for $x < x_{\mathsf{max}}$, else 1. Recommended: $\alpha = 0.75$, $x_{\mathsf{max}} = 100$.

GloVe vs. Word2Vec

GloVe	single-pass WLS over precomputed counts; faster with stored m
Word2Vec	online SGD; scales to streaming corpora

Similar embedding quality in practice.

Case Study: PCA on Financial Vocabulary

Project 30 financial terms into 2D via PCA on GloVe vectors. Colour by semantic group: rates & macro · equities & returns · risk · corporate actions & derivatives

Consistent patterns across pre-trained models

- *option*, *futures*, *derivative* $\rightarrow$ tight cluster
- *default*, *credit* $\rightarrow$ cluster together, away from equities
- *merger*, *acquisition* $\rightarrow$ nearly identical vectors
- *alpha*, *beta*, *factor* $\rightarrow$ cluster (shared asset-pricing origin)
- Rates cluster (*yield*, *spread*, *inflation*, *interest rate*) $\rightarrow$ most cohesive

Caution on axis interpretation

PCA axes $\mathbf{p}_1, \mathbf{p}_2$ are *not* interpretable as named financial concepts without further validation.

Full code: `code/notebooks/01-intro/demo.ipynb`